1.09 Quadratics: Problem Solving

Worked Solutions

1)

Length of pond: $40 - 2x$

Width of pond: $30 - 2x$

Area of pond is $30 \times 40 - 864 = 336\text{m}^2$

$(40 - 2x)(30 - 2x) = 336$

$4x^2 - 140x + 1200 = 336$

$4x^2 - 140x + 864 = 0$ so $(x - 8)(x - 27) = 0$

Gives $x = 8$ or $x = 27$

It can't physically be 27m so $x = 8$m

Length of pond: 24m

Width of pond: 14m

2)

$x^2 + 10(x + 2) = 44$

$\Rightarrow x^2 + 10x + 20 = 44$

$\Rightarrow x^2 + 10x - 24 = 0$

$\therefore x = -12$ or $x = 2$

It can't physically be -12 m so $x = 2$ m

3 a)

Minimum at $(-1, 2)$ so no real roots

3 b)

Maximum at $(-1, 2)$ so 2 real roots

3 c)

Minimum at $(4, -3)$ so 2 real roots

3 d)

Maximum at $(4, -3)$ so no real roots

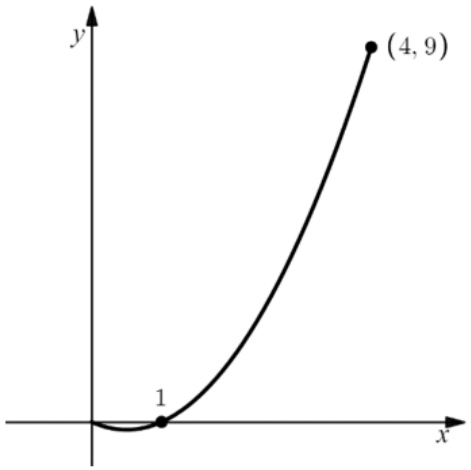
3 e)

Minimum at $(-2, 0)$ so 1 (repeated) real root

4 a)

When $x = 0, y = 0$, when $x = 4, y = 9$

When $y = 0, x = 0$ and $x = 1$

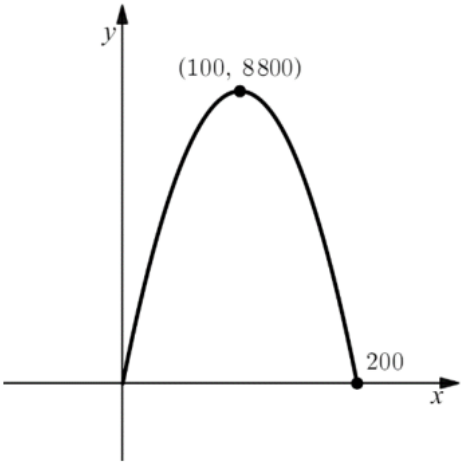


4 b)

Minimum at $x = \frac{1}{2}, y = -\frac{3}{16}$

5 a)

When $y = 0, x = 0$ and $x = 200$

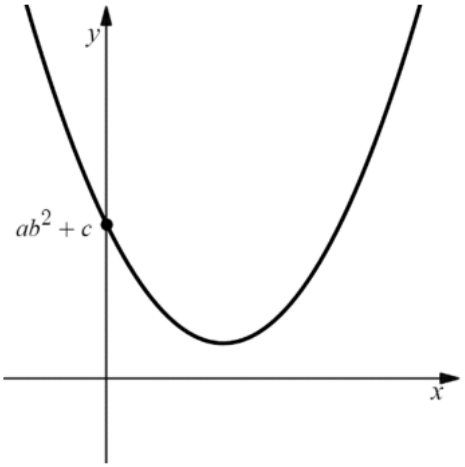


5 b)

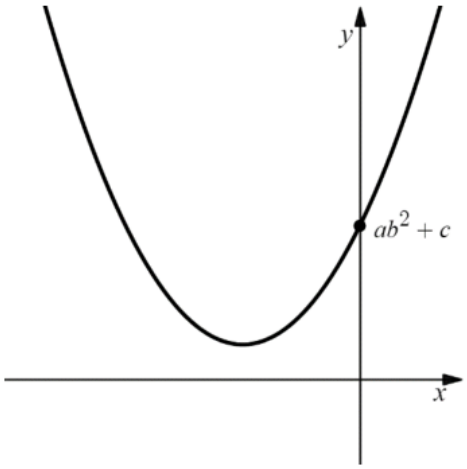
Maximum at $x = 100$,

$y = \frac{22}{25} \times 100 \times 100 = 8800$

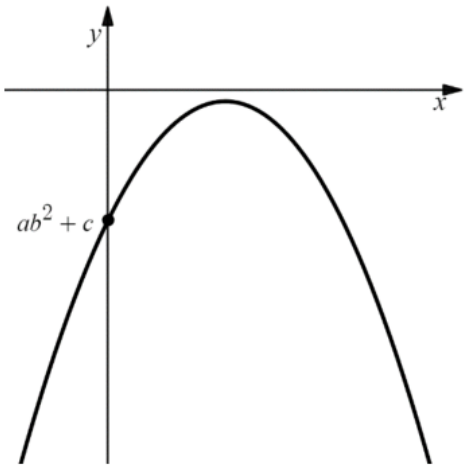
6 a)



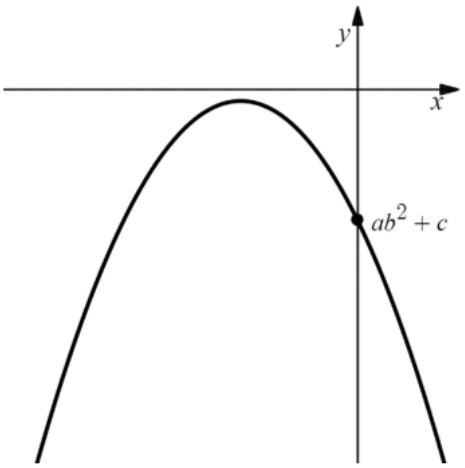
6 b)



6 c)



6 d)



7)

Minimum: $a > 0$

Minimum to have x-value < 0: $b < 0$

Top left quadrant: $c > 0$

$a > 0, b < 0$ and $c > 0$

8)

Maximum: $a < 0$

Maximum to have x-value > 0: $b > 0$

Maximum to have y-value > 0: $c > 0$

$a < 0, b > 0, c > 0$

y-intercept < 0:

$a(-b)^2 + c < 0$ so $ab^2 + c < 0$

$\therefore a < 0, b > 0, c > 0$ and $ab^2 + c < 0$

9 a)

$x^2 + 3x + k = (x + \frac{3}{2})^2 - (\frac{3}{2})^2 + k$

So $-(\frac{3}{2})^2 + k = -\frac{1}{2}$

$-\frac{9}{4} + k = -\frac{1}{2}$ so $k = \frac{7}{4}$

So it crosses the y-axis at $\frac{7}{4}$

$\therefore (0, \frac{7}{4})$

9 b)

$x^2 + 3x + \frac{7}{4} = 0 \Rightarrow x = -\frac{3}{2} \pm \frac{\sqrt{2}}{2}$

$\therefore (-\frac{3}{2} + \frac{\sqrt{2}}{2}, 0)$ and $(-\frac{3}{2} - \frac{\sqrt{2}}{2}, 0)$

10 a)

$-x^2 + 4kx + k = -(x-2)^2 + 5$

$= -(x^2 - 4x + 4) + 5 = -x^2 + 4x + 1$

so $k = 1$

So it crosses the y-axis at 1

$\therefore (0, 1)$

10 b)

$-x^2 + 4x + 1 = 0 \Rightarrow x = 2 \pm \sqrt{5}$

$\therefore (2 - \sqrt{5}, 0)$ and $(2 + \sqrt{5}, 0)$

11)

$$ax^2 + bx + c = 0$$
$$a(x^2 + \frac{b}{a}x + \frac{c}{a}) = 0$$
$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Complete the square:

$$(x + \frac{b}{2a})^2 - (\frac{b}{2a})^2 + \frac{c}{a} = 0$$
$$(x + \frac{b}{2a})^2 - \frac{b^2}{4a^2} + \frac{4ac}{4a^2} = 0$$
$$(x + \frac{b}{2a})^2 - \frac{b^2 - 4ac}{4a^2} = 0$$
$$(x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a^2}$$
$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$
$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ as req}$$

12)

$x = \frac{-1 \pm \sqrt{1-8k}}{4}$

So $\frac{-1-\sqrt{1-8k}}{4} + \frac{5}{2} = \frac{-1+\sqrt{1-8k}}{4}$

$\frac{9-\sqrt{1-8k}}{4} = \frac{-1+\sqrt{1-8k}}{4}$

$9 - \sqrt{1-8k} = -1 + \sqrt{1-8k}$

$10 = 2\sqrt{1-8k}$

$5 = \sqrt{1-8k}$

$25 = 1 - 8k$ so $k = -3$

13)

$x = \frac{k \pm \sqrt{k^2+60}}{-10}$

So $\frac{k-\sqrt{k^2+60}}{-10} + 1.6 = \frac{k+\sqrt{k^2+60}}{-10}$

$\frac{-16+k-\sqrt{k^2+60}}{-10} = \frac{k+\sqrt{k^2+60}}{-10}$

$-16 + k - \sqrt{k^2+60} = k + \sqrt{k^2+60}$

$-16 = 2\sqrt{k^2+60}$

$-8 = \sqrt{k^2+60}$

$64 = k^2 + 60$ so $k = \pm 2$